A Parameter-Space Study of Stellarator Coil Complexity Across Coil Types

What makes a stellarator hard to coil? Plasma-shape drivers across 500 quasi-isodynamic equilibria

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Motivation

Stellarator confinement is produced by complex 3D coils, and **coil complexity drives cost and feasibility**. A common assumption is that *high rotational transform (ι) devices are hard to coil with dipoles*.

With a large dataset we can test this directly: **which plasma-shape features actually make the coils hard to build?**

The scan

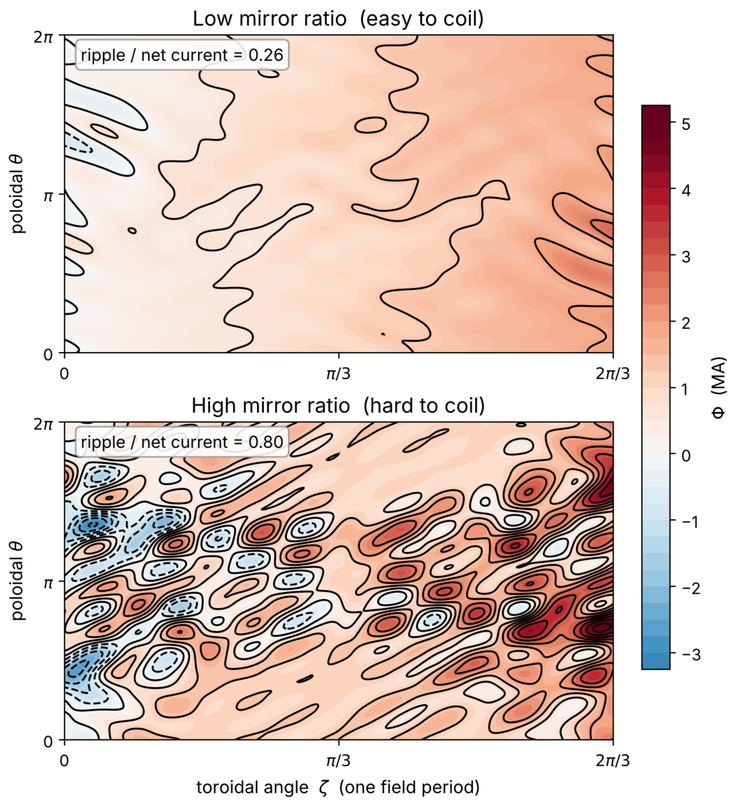
We run the **quadcoil** winding-surface optimizer over ∼500 quasi-isodynamic (QI, $N_{fp} = 3$) equilibria from the **ConStellaration** dataset. A current potential Φ on a winding surface sets the sheet current $\mathbf{K} = \mathbf{n} \times \nabla\Phi$; we solve for the field that reproduces the plasma boundary and read off a coil *cost* at a fixed field-error budget:

$$f_B = \frac{\oint (\mathbf{B}\cdot\mathbf{n})^2 dA}{\oint |\mathbf{B}|^2 dA}, \qquad \text{feasible} : f_B \le 1.01 f_B^{\text{target}}.$$

Six coil problems per configuration; *cost* is the objective achieved at the budget:

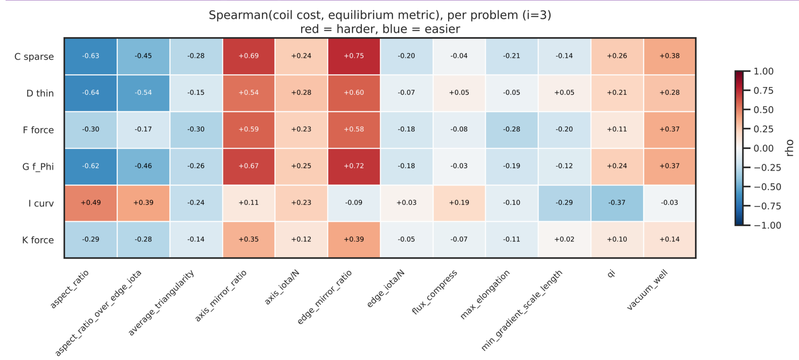
Problem	Objective	Penalizes
sparse dipole	$f_{\ell_1\Phi}$	dipole count (ℓ_1)
thin dipole	$f_{\max\Phi}$	peak current (ℓ_∞)
min-force dipole	$f_{\max F}$	peak coil force
low-curv filament	$f_{\max\kappa}$	filament curvature
min-force filament	$f_{\max F}$	peak coil force
L2 current pot.	f_Φ	smooth current (ℓ_2)

Coils from a current potential



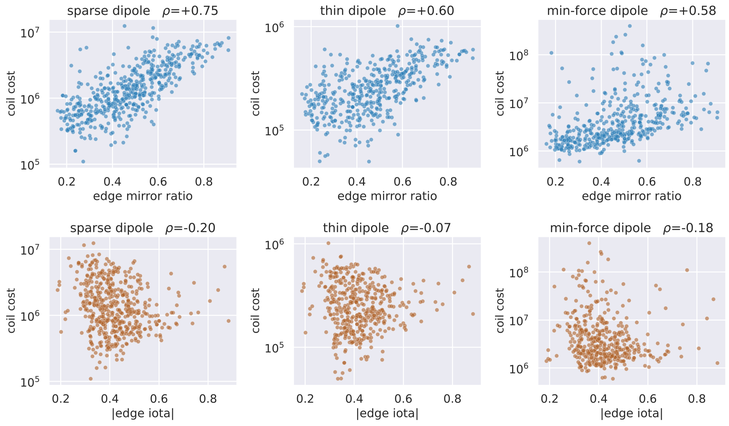
*Current potential Φ on the winding surface, one field period; **the black lines are the coils** (contours of Φ). The low-mirror case needs a few smooth coils; the high-mirror case needs many more, and they are contorted with closed windowpane islands. The single-valued ripple relative to the net current rises from* 0.26 *to* 0.80.

What drives coil complexity?



*Spearman ρ of coil cost vs each ConStellaration metric, per problem (red = harder, blue = easier). **Mirror ratio** (axis & edge) and **aspect ratio** are the strong, consistent signals; the two ι columns are weak and opposite in sign.*

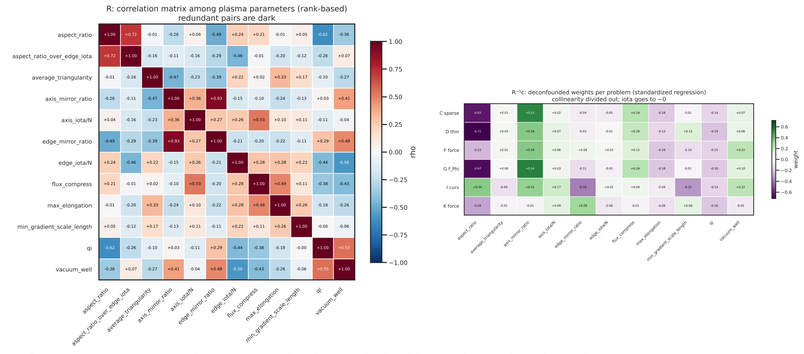
Mirror ratio, not rotational transform



Dipole cost rises steeply with mirror ratio ($\rho = +0.58$ to $+0.75$, top) but shows no clear trend with rotational transform (bottom), contrary to the common assumption.

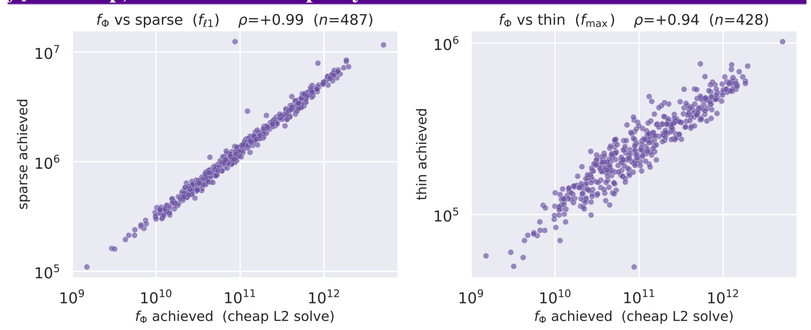
Is it causal, or just collinearity?

Plasma parameters are correlated among themselves, so a raw correlation can over-credit a correlated variable. The **coefficient of multiple correlation** isolates each feature's unique contribution: $R = \sqrt{\mathbf{c}^\top \mathbf{R}^{-1} \mathbf{c}}$, weights $\beta = \mathbf{R}^{-1}\mathbf{c}$ (**c**: cost–parameter correlations; **R**: correlations *among* parameters).



*Left: parameters are collinear (edge/axis mirror +0.93). Right: with collinearity divided out, **aspect ratio and axis mirror ratio remain**, while the ι weight collapses to ≈0. The driver is field-strength modulation and compactness, not rotational transform.*

f_Φ: a cheap, well-conditioned proxy



f_Φ (cheap L2 solve) vs the sparse and thin dipole costs; $n = 487$ and 428 (full ∼500 sweep).

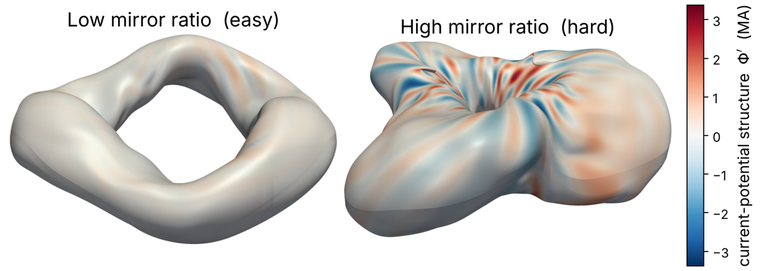
- A single well-conditioned linear solve.
- Tracks sparse / thin dipole cost: $\rho = +0.99/+0.94$.
- Smooth quadratic (good Hessian); ℓ_1/ℓ_∞ are smoothed but nearly flat, so Newton struggles.
- A cheap **difficulty screen** and candidate coil term for single-stage optimization.

Findings

- Mirror ratio and aspect ratio predict dipole coil complexity, **not rotational transform**.
- f_Φ is a cheap, well-conditioned proxy for the dipole-complexity objectives.

Scope. QI N_{fp}=3, current-potential formulation, one fixed f_B budget; these are associations (deconfounding controls measured metrics only).

The current potential in 3D



Current-potential structure Φ′ painted on the two winding surfaces (single-valued part, same scale). Low mirror ratio: smooth and nearly uniform, so the coils stay simple. High mirror ratio: strongly structured, so the coils become complex.

Next steps

- Confirm the f_Φ proxy across all f_B budgets (shown here at one budget, now on the full ∼500).
- Fold f_Φ into single-stage (plasma + coil) optimization as a cheap, differentiable coil term.
- Pin down the mechanism: *why* mirror ratio sets the required sheet-current structure.
- Extend beyond QI N_{fp}=3 (QA, QH, other field periods): does mirror + aspect generalize?
- Carry the drivers to cut filamentary coils with engineering limits (distance, HTS length); model *feasibility*, not just cost.

References & contact

QUADCOIL: Fu *et al.* · REGCOIL: Landreman & Boozer. · ConStellaration: Proxima Fusion (HuggingFace).

Contact: `lct9592@nyu.edu` · Kaptanoglu group, Courant Institute (NYU).

Code: `github.com/logantsai07/ConstellarationProject`

scan for code